

Junior Cloud Security Engineer

Internship Task Report

Intern Full Name	[REDACTED]
Intern Email	[REDACTED]
Cohort	Cohort 4, 2026
Task ID	CSE-AWS-L-010-04
Task Title	Secure Amazon S3 Bucket Configuration and Object-Level Data Protection
Skill Domain	AWS Cloud Security / IAM / S3 Access Control / Threat Detection / STS / IaC Security
Difficulty Level	Intermediate
CPE Credits	4 CPE Credits
Guiding Framework(s)	MITRE ATT&CK / NIST CSF / NIST SP 800-53 / CIS Benchmarks / ISO 27001 / OWASP / NICE Framework
Submission Date	29 Aug 2026

1.0 Task Overview

Task ID	CSE-AWS-L-010-04
Task Title	Secure Amazon S3 Bucket Configuration and Object-Level Data Protection
Skill Domain	AWS Cloud Security / Threat Detection / IAM / S3 Access Control
Difficulty	Intermediate
Duration	14 Hours
CPE Credits	4 CPE Credits
Cloud Platform(s)	AWS Free Tier
AWS Region(s) Used	<i>us-east-2</i>
Primary Tools	AWS CLI, AWS Management Console, AWS KMS, CloudShell, ChatGPT or Claude for AI assistance
Target Environment	<i>AWS Free Tier account / PwnedLabs lab / Cybr lab</i>
MITRE ATT&CK IDs	T1078 (Valid Accounts), T1530 (Data from Cloud Storage Object), T1098 (Account Manipulation)
Guiding Frameworks	<i>CIS AWS Foundations Benchmark v1.5, NIST SP 800-53, ISO 27001, OWASP Cloud Top 10</i>
Regulatory Alignment	<i>ISO 27001 A.9 Access Control, A.12 Operations Security; NICE PR-CDA-001; SFIA Level 3-4</i>

1.1 Introduction

This task focused on securing Amazon S3 storage against common cloud security risks such as public data exposure, excessive permissions, and insufficient logging. The lab demonstrated how encryption, least-privilege access controls, private object permissions, and server access logging can be combined to protect sensitive data and improve visibility into S3 activity.

1.2 Objectives & Learning Outcomes

- **Understand S3 security risks and access-control mechanisms** — Reviewed common S3 misconfigurations, including public access, insecure ACLs, and overly permissive policies, along with appropriate mitigations.
- **Configure a secure S3 environment** — Created an S3 lab bucket, enabled **Block Public Access and versioning**, and configured a customer-managed AWS KMS key for encryption testing.
- **Implement encryption and least-privilege access** — Uploaded confidential.txt using **SSE-KMS** with S3LabKey and configured permissions to restrict access to the authorized IAM user.
- **Apply object-level security and audit logging** — Configured private object permissions and enabled **S3 Server Access Logging** to a separate private audit bucket for security monitoring and auditability.
- **Validate and document S3 security controls** — Tested anonymous access to confirm that unauthorized access was denied, reviewed generated access logs, assessed identified risks using a risk matrix, and documented the results in the final security report.

1.2 Scope & Legal Authorisation

All activities performed during this task were conducted within **authorised AWS lab environments** created specifically for security testing and educational purposes. Testing was limited to resources under the student's control, and no unauthorised third-party or production systems were accessed.

Authorised environments used:

- **Personal AWS lab account** — Used to create and configure S3 buckets, IAM permissions, AWS KMS encryption keys, and security controls.
- **secure-s3-lab-sk S3 bucket** — Authorised source bucket used for encryption, access-control, ACL, and read/write testing.
- **secure-s3-audit S3 bucket** — Private audit bucket used to receive and review S3 server access logs.
- **AWS IAM and AWS KMS** — Used within the same authorised account to configure least-privilege permissions and the S3LabKey encryption key.

1.4 MITRE ATT&CK Cloud Threat Mapping

MITRE ATT&CK Technique ID	Technique Name	Relevance to This Task
T1530	Data from Cloud Storage	The lab evaluated the risk of unauthorized access to data stored in S3 due to public access, insecure ACLs, or overly permissive policies. Private permissions and Block Public Access were used to mitigate this risk.
T1078	Valid Accounts	The lab used an IAM user ([REDACTED]) to access S3 and KMS resources, demonstrating why permissions assigned to valid cloud identities must follow least-privilege principles.
T1562.008	Impair Defenses: Disable or Modify Cloud Logs	S3 server access logging was enabled and logs we

2.0 Environment & Tool Setup

2.1 Cloud Architecture Overview

Cloud Resource / Component	Type / Service	Role in Task	Region / Config Notes
secure-s3-lab-sk	Amazon S3	Primary storage bucket used for uploading and securing confidential.txt	Private / Block Public Access enabled / Versioning enabled
secure-s3-audit	Amazon S3	Separate destination for S3 server access logs	Private / Block Public Access enabled / logs/ prefix
S3LabKey	AWS KMS	Customer-managed key used to encrypt S3 objects with SSE-KMS	Used with alias/S3LabKey
[REDACTED]	AWS IAM User	Authorized lab identity used to perform S3 and KMS operations	Least-privilege S3/KMS permissions configured
AWS CLI	Local/CloudShell Tool	Used to upload encrypted objects, modify ACLs, verify encryption, and test S3 access	Authenticated as [REDACTED]

2.2 AWS CLI & Tool Configuration

Tool / Service	Version	Installation Method	Purpose in This Task
AWS CLI	AWS CLI v2	Pre-installed in AWS CloudShell	Executed S3, IAM, and KMS commands; uploaded encrypted objects; modified ACLs; and verified security configurations
Amazon S3	AWS Managed Service	AWS Management Console — no installation required	Created and configured the primary S3 bucket and audit bucket; enabled versioning, Block Public Access, and server access logging
AWS IAM	AWS Managed Service	AWS Management Console — no installation required	Configured permissions for the [REDACTED] IAM user and implemented least-privilege access to S3 and KMS resources
AWS KMS	AWS Managed Service	AWS Management Console — no installation required	Created and managed S3LabKey and provided SSE-KMS encryption for confidential.txt
AWS CloudShell	AWS Managed Service	Launched through AWS Management Console — no local installation required	Provided the command-line environment used to execute AWS CLI commands and capture lab evidence

Key Configuration Commands Used

The following AWS CLI commands were used to verify the authenticated IAM identity, confirm access to the required AWS services, and validate the lab environment before executing the security tests.

1. Verify AWS CLI identity

```
aws sts get-caller-identity
```

Actual output:

```
{
  "UserId": "[REDACTED]",
  "Account": "[REDACTED]",
  "Arn": "[REDACTED]:[REDACTED]:[REDACTED]"
}
```

This confirmed that AWS CLI commands were being executed using the authorized [REDACTED] IAM user.

2. Verify available S3 buckets

```
aws s3 ls
```

This command was used to verify S3 connectivity and confirm access to the lab buckets, including:

```
secure-s3-lab-sk
```

```
secure-s3-audit
```

3. Verify KMS key alias

```
aws kms list-aliases
```

The output was reviewed to confirm that the encryption key alias existed:

```
alias/S3LabKey
```

This confirmed that the KMS key required for the SSE-KMS encryption test was available.

4. Verify source bucket configuration

```
aws s3api get-bucket-versioning \
```

```
--bucket secure-s3-lab-sk
```

Expected/observed configuration:

```
{
  "Status": "Enabled"
}
```

This verified that **S3 Versioning** was enabled on the lab bucket.

5. Verify Block Public Access

```
aws s3api get-public-access-block \
```

```
--bucket secure-s3-lab-sk
```

The configuration should show the four public-access protection settings as true, confirming that **Block Public Access** was enabled.

2.3 AI Tool Usage Declaration

AI Tool	Used For	What Was NOT Done with AI
ChatGPT	Explaining S3 security concepts, clarifying SSE-S3 vs. SSE-KMS, explaining IAM/bucket policy/ACL concepts, troubleshooting AWS permission errors, summarising observations, and assisting with report wording and risk analysis.	AI did not directly access the AWS account, execute AWS CLI commands, perform the lab activities, capture screenshots, or generate technical evidence. All configurations, tests, CLI outputs, and screenshots were produced by the intern in the authorised AWS lab environment.

3.0 Methodology — Phase Execution

Phase A: Theory & Conceptual Foundation

Phase Objective	Understand the security risks of misconfigured S3 buckets and explore how encryption and access control mitigate these risks.
Estimated Time	~90 minutes
Evidence File(s)	PhaseA_Summary.pdf

Action Taken

Reviewed common Amazon S3 security misconfigurations, including public read/write access, overly permissive bucket policies, and insecure ACL configurations. Studied **SSE-S3** and **SSE-KMS encryption** and reviewed how **IAM roles**, **bucket policies**, and **object ACLs** work together to control access to S3 resources.

Key Observations

- **Public read access** can expose sensitive data to unauthorized users.
- **Public write access** can allow attackers to upload, modify, or delete objects.
- **SSE-S3** uses encryption keys managed automatically by AWS.
- **SSE-KMS** provides greater control through AWS KMS, including key policies, permissions, and auditing.
- **IAM roles** define what an AWS identity can do.

- **Bucket policies** provide resource-based permissions at the S3 bucket/object level.
- **Object ACLs** are older object-level access controls and are generally less preferred than IAM and bucket policies.
- Applying **least privilege, S3 Block Public Access, encryption, and regular permission reviews** significantly reduces S3 exposure.

Bucket Policy vs. Object ACL

A **bucket policy** is a JSON-based resource policy attached to an S3 bucket that can control access to the bucket and its objects for multiple users or accounts. An **object ACL** is attached to an individual object and grants basic permissions such as read or write access to specific AWS accounts or predefined groups.

Evidence

PhaseA_Summary.pdf — One-page summary containing examples of S3 misconfigurations, associated security risks, SSE-S3 vs. SSE-KMS comparison, access-control concepts, and recommended mitigations.

Phase B: Cloud Environment Setup & Target Preparation

Phase Objective	Create a secure testing environment in your own AWS account.
Estimated Time	~120 minutes
Evidence File(s)	PhaseB_config(1).png , phaseB_config(2).png

Actions Taken

Created a dedicated **Amazon S3 testing environment** in the AWS account. Created the secure-s3-lab bucket, enabled **Block All Public Access**, enabled **bucket versioning**, and created a KMS key named S3LabKey for encryption testing.

Key Observations

- **Block Public Access** protects the bucket from accidental public exposure.
- **Versioning** maintains previous versions of objects, helping with recovery from accidental modification or deletion.
- **AWS KMS** provides controlled encryption-key management and allows key usage to be audited.
- Combining access restrictions, versioning, and encryption establishes a safer environment for subsequent S3 security testing.

Evidence

PhaseB_Config(1).png — Screenshot of the S3 bucket configuration showing **Block Public Access enabled** and **Bucket Versioning enabled**.

PhaseB_Config(2).png - Bucket kMS

Phase C: Core Execution

Phase Objective	Apply encryption at rest using SSE-KMS and enforce least privilege through IAM policies.
Estimated Time	~150 minutes
Evidence File(s)	PhaseC_Encryption.log, PhaseC_Anonymous_Access_Denied.png

Actions Taken

Configured Amazon S3 server-side encryption using AWS KMS (SSE-KMS) and uploaded confidential.txt to the secure-s3-lab-sk bucket using the customer-managed key alias/S3LabKey. Verified encryption through AWS CLI, applied a bucket policy granting access only to the IAM user [REDACTED], and validated that anonymous access to the object was denied.

Key Observations

- The uploaded object was encrypted using SSE-KMS (aws:kms).
- The bucket policy enforced least privilege by allowing only the [REDACTED] IAM user access.
- Anonymous browser access returned AccessDenied, confirming that the bucket remained private despite the object being uploaded.

Evidence Produced:

PhaseC_Encryption.log: successful SSE-KMS encryption/upload evidence

PhaseC_Anonymous_Access_Denied.png — failed anonymous browser access

Phase D: Independent Extension & Advanced Implementation

Phase Objective	Extend security by applying object ACLs and enabling logging for auditability.
Estimated Time	~90 minutes
Evidence File(s)	PhaseD_Logs.png

Actions Taken

Modified the ACL of confidential.txt to enforce private object-level access. Created a separate private S3 audit bucket and enabled server access logging on the primary secure-s3-lab-sk bucket. Performed read, write, and listing operations against the source bucket to generate auditable activity and verified that S3 log objects were delivered to the audit bucket.

Key Observations: Setting the object ACL to private prevents ACL-based public access to confidential.txt. Server access logging provides an audit trail of requests made against the source S3 bucket. Separating application data and

audit logs into different private buckets improves security and makes activity easier to review. Log delivery is asynchronous, so audit files may not appear immediately after testing.

Evidence: PhaseD_Logs.png — Screenshot of the private audit S3 bucket showing server access log files generated after read/write operations were performed against secure-s3-lab-sk.

Risk Matrix (Phase D Output)

Cloud Asset / Resource	Threat / Risk	MITRE Ref.	Likelihood (1–5)	Impact (1–5)	Score	Rating
S3 Bucket (secure-s3-lab-sk)	Public exposure caused by incorrect ACL or access configuration	T1530	3	5	15	Critical
S3 Object (confidential.txt)	Unauthorized access to sensitive data due to excessive object permissions	T1530	3	4	12	High
S3 Audit Logging	Disabled or missing access logging reduces ability to detect and investigate unauthorized S3 activity	T1562.008	3	4	12	High
KMS Key (S3LabKey)	Excessive KMS permissions could allow unauthorized use of encryption/decryption keys	T1098	2	5	10	High

Phase E: Hosted Lab Integration

Phase Objective	<i>[State the objective of this phase as written in your task brief]</i>
Estimated Time	<i>[~90–120 minutes]</i>
Evidence File(s)	<i>[Filename(s) produced — e.g. PhaseA_Notes.pdf]</i>

Actions Taken

[Describe step-by-step what you did in this phase. Be specific — tool commands run, configurations applied, data examined, or concepts studied. Do not copy the task instructions verbatim; write what you actually did and observed.]

Include relevant command outputs, configuration snippets, or tool screenshots inline. Label each screenshot as Figure X with a one-sentence caption.

Key Observations

[What did you notice, discover, or conclude during this phase? This is your analytical voice — not a list of steps but what the steps revealed.]

Evidence Produced

[List the evidence files, screenshots, or artefacts you saved during this phase. State the filename and a one-line description of its contents.]

- [e.g. PhaseA_Notes.pdf — annotated summary of OWASP ASVS control mapping with comparison table]
- [Add further evidence items as needed]

Phase F: Documentation & Reporting

Phase Objective	Compile findings into a professional report demonstrating secure configuration practices.
Estimated Time	[~60–90 minutes]
Evidence File(s)	CSE_CSE-AWS-L-010-04_20260829.pdf

Actions Taken

Compiled all S3 lab activities, configurations, security findings, risk analysis, and supporting evidence into a structured final security report. Documented the controls implemented for encryption, access management, object permissions, and logging.

Key Observations:

The completed lab demonstrated that combining **SSE-KMS encryption, least-privilege access, private object permissions, Block Public Access, and server access logging** provides stronger protection for S3 data. Proper configuration and continuous auditing are important for preventing unauthorized access and identifying security events.

Evidence Required:

CSE_CSE-AWS-L-010-04_20260829.pdf— Final professional report summarizing the S3 security assessment, implemented controls, key findings, risk matrix, supporting screenshots, and security recommendations.

4.0 Findings

4.1 Findings Summary

#	Finding Title	Type / Category	Cloud Service / Resource	MITRE Ref.	Phase	Severity	Evidence Ref.
1	Risk of Public S3 Data Exposure	Access Control Weakness	Amazon S3 / secure-s3-lab-sk	T1530	C	Critical	PhaseC_Anonymous_Access_Denied.png
2	Excessive IAM/KMS Permissions	IAM Policy Gap	IAM [REDACTED] / KMS S3LabKey	T1078	C	High	PhaseC_Encryption.log
3	Insufficient S3 Audit Visibility	Logging Gap	Amazon S3 / secure-s3-audit	T1562.008	D	High	PhaseD_Logs.png
4	Object-Level Permission Exposure	Access Control Weakness	S3 Object / confidential.txt	T1530	D	High	Object ACL CLI output / Phase D evidence

4.2 Detailed Finding Analysis

Finding 1: Risk of Public S3 Data Exposure

Field	Content
Finding Title	Risk of Public S3 Data Exposure
Type / Category	Access Control Weakness / Misconfiguration
Cloud Service / Resource	Amazon S3 / secure-s3-lab-sk / confidential.txt
Description	Public or overly permissive S3 permissions could expose sensitive objects to unauthorized users. The lab evaluated this risk by configuring Block Public Access , restricting bucket access to the authorized IAM user, and testing anonymous access to confidential.txt. The anonymous browser request returned AccessDenied , confirming that the implemented controls prevented public access.
How Identified	Reviewed the S3 bucket permissions and performed an anonymous browser-access test against confidential.txt. The request returned AccessDenied , validating that the object was not publicly accessible.
MITRE ATT&CK Reference	T1530 – Data from Cloud Storage
Business / Security Impact	If an S3 bucket or object were accidentally made public, an unauthorized user could retrieve sensitive information. This could result in loss of confidentiality, data exposure, regulatory concerns, and reputational impact.
Severity / Impact	Critical
Evidence Reference	PhaseC_Anonymous_Access_Denied.png

Finding 2: Excessive IAM/KMS Permissions

Field	Content
Finding Title	Excessive IAM/KMS Permissions
Type / Category	IAM Policy Gap / Access Control Weakness

Field	Content
Cloud Service / Resource	AWS IAM user [REDACTED] / AWS KMS key S3LabKey
Description	During the lab, the [REDACTED] IAM user required permissions to perform S3 operations and use the KMS key for SSE-KMS encryption. Permission errors demonstrated the dependency between IAM permissions and KMS key access. Excessively broad S3 or KMS permissions could allow an identity to access, modify, encrypt, or decrypt resources beyond its intended responsibilities. Least-privilege permissions were therefore applied for the required lab operations.
How Identified	Identified through AWS CLI testing and AccessDenied responses while performing S3 and KMS operations. Commands such as aws kms list-aliases, aws s3 cp, and aws sts get-caller-identity were used to validate the identity and required permissions.
MITRE ATT&CK Reference	T1078 – Valid Accounts
Business / Security Impact	Excessive permissions assigned to a valid IAM identity could increase the impact of compromised credentials. An attacker using the account could potentially access sensitive S3 objects or misuse KMS capabilities, affecting data confidentiality and integrity.
Severity / Impact	High
Evidence Reference	PhaseC_Encryption.log / IAM permission testing CLI output

Finding 3: Insufficient S3 Audit Visibility

Field	Content
Finding Title	Insufficient S3 Audit Visibility
Type / Category	Logging Gap
Cloud Service / Resource	Amazon S3 / secure-s3-lab-sk and secure-s3-audit
Description	Without server access logging, requests made against the primary S3 bucket would have reduced audit visibility. Server access logging was enabled on secure-s3-lab-sk, with generated logs delivered to the separate private secure-s3-audit bucket under the logs/ prefix.
How Identified	Reviewed the Server access logging configuration in the AWS S3 Console and performed read/write operations against the source bucket. Log generation was then validated by reviewing objects delivered to the audit bucket.
MITRE ATT&CK Reference	T1562.008 – Impair Defenses: Disable or Modify Cloud Logs

Field	Content
Business / Security Impact	Missing or disabled logging could reduce the organization's ability to detect suspicious S3 activity, investigate unauthorized access, and maintain an audit trail. This could delay incident response and affect security monitoring and compliance activities.
Severity / Impact	High
Evidence Reference	PhaseD_Logs.png

5.0 Risk Assessment & Cloud Threat Model

Finding / Gap	Observable / Trigger in Cloud Telemetry	Recommended Control	AWS Service / Tool for Implementation
Finding 1 – Risk of Public S3 Data Exposure	Public bucket-policy/ACL changes, anonymous GetObject attempts, or unexpected access to S3 objects	Enable S3 Block Public Access , maintain restrictive bucket policies, disable unnecessary ACLs, and continuously check for externally accessible resources	Amazon S3 Block Public Access / IAM Access Analyzer / AWS Config
Finding 2 – Excessive IAM/KMS Permissions	IAM policy changes, KMS permission changes, or unexpected use of KMS keys by authorized identities	Apply least-privilege IAM and KMS policies , restrict permissions to required S3/KMS actions, and periodically review unused or excessive permissions	AWS IAM / AWS KMS / IAM Access Analyzer / AWS CloudTrail
Finding 3 – Insufficient S3 Audit Visibility	Missing S3 access-log delivery, changes to logging configuration, or unexpected S3 read/write activity	Enable S3 Server Access Logging , store logs in a separate private audit bucket, restrict modification/deletion of logs, and monitor logging configuration	Amazon S3 Server Access Logging / AWS Config / AWS CloudTrail / Amazon CloudWatch

5.2 Risk Exposure Over Time

Risk 1 — Risk of Public S3 Data Exposure:

If S3 public-access controls or bucket permissions become misconfigured, objects such as confidential.txt could become accessible to unauthorized users. If left undetected, this could result in prolonged data exposure or exfiltration and create confidentiality and compliance risks. The lab's successful **AccessDenied** test demonstrated why Block Public Access and restrictive permissions should remain enforced.

Risk 2 — Excessive IAM/KMS Permissions:

If the [REDACTED] account retains more S3 or KMS permissions than necessary, compromised credentials could provide an attacker with greater access to stored and encrypted data. Over time, unused or excessive permissions can increase the potential impact of account compromise, including unauthorized object access or misuse of S3LabKey. Regular permission reviews and least-privilege policies reduce this exposure.

Risk 3 — Insufficient S3 Audit Visibility:

If server access logging is disabled or log delivery to secure-s3-audit stops, S3 activity may occur without sufficient records for investigation. Prolonged logging gaps could make unauthorized reads, writes, or other suspicious activity harder to identify and reconstruct during an incident. Maintaining a separate protected audit bucket preserves visibility and supports security investigations.

6.0 Recommendations

[All recommendations must be grounded in your actual findings from Section 4. Generic cloud security advice not tied to a specific finding is not acceptable. Every recommendation should trace back to a finding in the summary table.]

6.1 Technical Remediation

#	Finding	Specific Remediation Action	Priority	Standard / Reference
1	Risk of Public S3 Data Exposure	Keep S3 Block Public Access enabled on secure-s3-lab-sk, maintain restrictive bucket policies, and keep objects private. Verify using <code>aws s3api get-public-access-block --bucket secure-s3-lab-sk</code> . Periodically use IAM Access Analyzer to identify unintended external access.	Immediate	CIS AWS Foundations / NIST SP 800-53 AC-3 / ISO 27001 Access Control
2	Excessive IAM/KMS Permissions	Review the [REDACTED] IAM policy and remove permissions not required for the lab. Restrict S3 permissions to secure-s3-lab-sk and restrict KMS permissions to S3LabKey rather than using "Resource": "*", where supported. Regularly review permissions with IAM Access Analyzer.	Short-term	NIST SP 800-53 AC-6 (Least Privilege) / ISO 27001 Access Control
3	Insufficient S3 Audit Visibility	Keep S3 Server Access Logging enabled for secure-s3-lab-sk with logs delivered to the private secure-s3-audit bucket. Restrict access to the audit bucket and periodically verify that new log objects are being generated under logs/.	Immediate	NIST SP 800-53 AU-2/AU-6 / ISO 27001 Logging & Monitoring

6.2 Strategic & Operational Improvements

- **Enforce S3 Block Public Access at the account level** to prevent buckets and objects from being accidentally exposed. Use AWS Config or IAM Access Analyzer to continuously identify buckets with unintended public or cross-account access.
- **Establish periodic IAM and KMS permission reviews** to ensure users such as [REDACTED] retain only the permissions required for their responsibilities. Remove unused S3/KMS actions and avoid wildcard permissions wherever practical.
- **Centralize and protect S3 security logging** by sending access logs to a dedicated private audit bucket such as secure-s3-audit. Restrict modification and deletion of audit data and periodically verify that logging remains operational.
- **Standardize encryption requirements for sensitive S3 data** by requiring server-side encryption and using SSE-KMS where stronger key control and auditing are required. KMS key policies should also follow least-privilege principles.
- **Implement continuous configuration monitoring** using AWS Config, Security Hub, CloudTrail, and CloudWatch where appropriate to detect changes to bucket policies, public-access settings, encryption, logging, and other security-sensitive S3 configurations.

7.0 Reflection

1. How does enabling versioning contribute to data resilience during ransomware attacks?
S3 Versioning preserves previous versions of objects when files are modified or deleted. During a ransomware attack, if objects are encrypted, overwritten, or deleted by an attacker, earlier versions may still be available for recovery. Versioning therefore improves resilience and reduces the risk of permanent data loss, especially when combined with strong access controls and protected backups.
2. What business risks arise if an organization fails to encrypt sensitive data stored in S3?
Failure to encrypt sensitive S3 data increases the impact of unauthorized access or data exposure. It can result in loss of confidentiality, regulatory or contractual non-compliance, financial penalties, reputational damage, and loss of customer trust. Using encryption such as **SSE-KMS** also provides stronger control and auditing of encryption-key usage.
3. How would you design a monitoring strategy for detecting unauthorized object downloads?
I would enable **AWS CloudTrail S3 data events** for important buckets to record object-level API activity such as GetObject. Logs could be monitored using CloudWatch and alerts generated for suspicious behavior, such as unusual download volumes, unexpected identities, or access from unfamiliar sources. **S3 server access logging, GuardDuty, and Security Hub** can provide additional visibility and detection capabilities.
4. If you discovered a misconfigured public bucket belonging to another company, what ethical steps should you take?
I would **not attempt to download, modify, delete, or further explore the data** because I do not have authorization. I would document only the minimum information necessary to describe the apparent exposure and report it through the organization's official security or vulnerability-disclosure channel. I would keep the information confidential and allow the organization to investigate and remediate the issue rather than attempting additional testing without explicit authorization.

8.0 Deliverables Submission Checklist

✓	Phase	Deliverable Description	Your Filename	Status
☐	Phase A	Theory evidence — e.g. architecture/relationship diagram, notes PDF, AI summary text file, annotated MITRE mapping	PhaseA_Summary.pdf	<i>Complete</i>
☐	Phase B	Environment setup evidence — e.g. VPC dashboard screenshot, CloudTrail config proof, IAM policy JSON, get-caller-identity output, tool version screenshots	PhaseB_config(1).png , phaseB_config(2).png	<i>Complete</i>
☐	Phase C	Core execution evidence — e.g. CLI output log file, query results screenshot, policy test JSON, assume-role output, Athena query result, GuardDuty finding screenshot	PhaseC_Encryption.log, PhaseC_Anonymous_Access_Denied.png	<i>Complete</i>
☐	Phase D	Extension & risk model evidence — e.g. hardened policy JSON, risk matrix XLSX/screenshot, threat model diagram, hosted lab completion badge, AI analysis summary	PhaseD_Logs.png	<i>Complete</i>
☐	Phase E	Hosted lab completion — e.g. TryHackMe badge screenshot, PwnedLabs completion proof, Cybr lab certificate (if required by your task brief)	<i>[Your filename]</i>	<i>Pending</i>
☐	Report	This report — exported as PDF using the naming convention below	CSE_CSE-AWS-L-010-04_20260829.pdf	<i>Complete</i>

9.0 Additional Observations

- S3 security is most effective when **encryption, access control, versioning, and logging** are implemented together rather than relying on a single control.
- **SSE-KMS** provides additional control and auditability compared with basic S3-managed encryption, but IAM identities must also have appropriate KMS permissions.
- **Least-privilege IAM permissions** can require careful testing because S3 bucket-level, object-level, and KMS actions require different permissions.
- **Block Public Access** provides an important safeguard against accidental exposure even when bucket policies or ACLs are modified.
- **S3 Versioning** improves recovery options following accidental deletion, modification, or malicious activity.
- Using a **separate private audit bucket** helps preserve security logs and separates operational data from audit records.
- S3 server access logs are **not delivered immediately**, so monitoring processes should account for delivery delays.
- Regular permission reviews and configuration monitoring are necessary because a secure initial configuration can become insecure through later policy or access-control changes.

Declaration & Sign-Off

I confirm that:

- All work in this report is my own except where AI usage is explicitly declared in Section 2.3.
- All technical activities were performed exclusively within the authorised cloud environments listed in Section 1.3.
- No external AWS accounts, production environments, third-party cloud resources, or real customer data were accessed during this task.
- All AWS credentials, access keys, session tokens, and account identifiers used during this task have been redacted in evidence files before submission.
- I have reviewed this report for accuracy, completeness, and professional presentation before submission.

Intern Full Name	Signature	Date of Submission
[REDACTED]	[REDACTED]	29 Aug 2026